Emma-Caroline Avery

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EDUCATION

Barnard College of Columbia University | New York, NY | *B.A. in Psychology* Expected May 2027

BASEBALL & PLAYER DEVELOPMENT EXPERIENCE

Columbia University Baseball & SABRLions | New York, NY Jan. 2024 - Present

Director of Analytics, Columbia Baseball (coaching staff) Sep. 2026 - Present

- Serve on the Columbia Baseball staff as the program's point person for Trackman data, reporting, and player-development analytics
- Manage the team's Trackman data pipeline: operate the stadium and portable units with synchronized iPad video, set up Edgertronic cameras, clean, verify, and retag pitch data, and maintain the season database behind every report
- Write player-development reports tracking pitcher progress and review video in TruMedia and game footage on AWRE, translating findings for coaches and players; wrote advance-scouting reports on opponents in earlier seasons
- Write per-pitcher outing notes for the pitching coach, and built a one-page 2025-26 D1 pitch-characteristics reference for the staff

Director, SABRLions Aug. 2026 - Present

- Lead Columbia Baseball's student-run analytics and operations group: set priorities, run weekly meetings, recruit and onboard new members, and coordinate guest speakers from professional baseball
- Oversee game-day technology operations, assigning members to Trackman shifts across practices, scrimmages, and games; train new members with two browser-based trainers I built (a live-tagging simulator and a pitch-recognition game) and wrote the onboarding guide
- Run SABRLions' social accounts; published *Building a Trackman Team: Lessons from the SABRLions* (Medium, May 2026)

Also: Director of Game Day Technology (Jan. 2026 - Present); Student Manager (Jan. 2024 - Present)

Paradigm Player Development Solutions | Phoenix, AZ | *Strategy & Solutions Analyst* Sep. 2026 - Present

- Consultant for two of Paradigm's client programs: review their players' performance data, evaluate the analytics they receive, and deliver feedback and development suggestions to the coaching staff
- Own intake and turnaround of ad-hoc coach requests, walk coaching staffs through Paradigm reports and tools, and capture coach feedback in writing so it reaches the analysts and product team
- Analyzed a client program's Blast Motion bat-sensor data and wrote the staff-facing report: which swing metrics hold up between sessions, how large a change has to be to beat session-to-session noise, and D1 reference ranges for each metric
- Research and write threads for Paradigm's X account, including how long college stats take to become reliable (6.56M D1 pitches, 2022-26) and what Blast swing data shows about rotational vs. linear swings

Los Angeles Dodgers | Glendale, AZ | *Player Development Intern* May 2026 - Aug. 2026

- Worked across ACL hitting, pitching, catching, performance science, biomechanics, and mental performance; joined position-group, player-plan, staff, and GM meetings
- Operated KinaTrax, Trackman, HitTrax, and Trajekt during MLB and MiLB bullpens, live BP, cage work, rehab, testing, and pitch design
- Analyzed KinaTrax biomechanical data to identify player trends, translating findings into staff-facing reports and presentations; wrote a season-long ACL player-development report and helped build individualized player and goal plans with coaches
- Co-designed and built a custom performance-science testing apparatus for a hitter-movement study; captured and analyzed data and presented findings to staff

Oakland Athletics | Oakland, CA | *Baseball Operations & Analytics Intern* Jun. 2024 - Jul. 2024

- Conducted data cleaning and analysis, focusing on overlooked players in the minor leagues for the trade deadline, and supported pre-draft analysis for the 2024 MLB Draft
- Operated KinaTrax during home games to collect biomechanical player-performance data for organizational use

INDEPENDENT PROJECTS (for my own development and to support Columbia Baseball)

- **Video pipelines:** pull Trackman video through its API and FTP and AWRE game footage through the AWRE API, cut it into per-pitch clips synced to Trackman data, and organize 20,000+ clips by pitcher and pitch
- **Pose estimation:** run skeleton tracking on bullpen iPad, Edgertronic, and AWRE game video (~6,500 pitches) to build delivery metrics, overlays, per-pitcher delivery cards, release-frame detection, and time to plate
- **Pitch tipping:** tested set-position and delivery video for tells that predict pitch type, with cross-validation and multiple-testing correction, and made side-by-side comparison videos for flagged pitchers
- **Research:** arm slot, usage, and availability across ~17,000 D1 pitcher-seasons (Medium article in progress); bullpen-to-game transfer; pitch-quality (Stuff+) modeling; mental performance signals in Trackman data

TECHNICAL SKILLS & LANGUAGES

Baseball technology: Trackman, KinaTrax, Hawkeye, Rapsodo, HitTrax, Trajekt, Blast Motion, Qualisys, Edgertronic, AWRE, Sony, Synergy, TruMedia; familiar with BATS

Analysis and tools: Microsoft Excel, R, Python (pose estimation with RTMPose/YOLO, Trackman and AWRE APIs, FTP pipelines), SQL (developing), Adobe Lightroom, Adobe Premiere Pro, Canva

Languages: Spanish (Intermediate), Italian (Elementary)

Certifications: Human Subjects Protection SBR and Research with Minors SBR (September 2024); Rapsodo Pitching Certification (in progress)